

MARTE in UML

MARTE stands for:

Modeling and Analysis of Real-Time and Embedded systems

It is a standardized UML profile defined by the Object Management Group (OMG) to extend UML for:

- Real-time systems
- Embedded systems
- Performance-critical systems

Why MARTE was Introduced

Standard UML lacks precise constructs for:

- Timing constraints (deadlines, latency)
- Resource modeling (CPU, memory, hardware)
- Concurrency and parallelism
- Performance analysis
- Embedded system architecture

MARTE addresses these gaps using:

- Stereotypes
- Tagged values
- Constraints

(A) Modeling Real-Time Systems

- Systems where correctness depends on time constraints
- Example: automotive control systems, avionics

(B) Modeling Embedded Systems

- Systems tightly coupled with hardware
- Example: IoT devices, mobile firmware

(C) Quantitative Analysis

- Performance (latency, throughput)
- Scheduling
- Resource utilization

3.1 Time Modeling

- Logical time vs physical time
- Clocks and time units
- Timed events

Example: Execution must complete within 10 ms

Represents both hardware and software resources:

- CPU
- Memory
- Communication buses
- Threads/processes

Stereotypes (Extension Mechanism)

Examples of MARTE stereotypes:

- <<RtUnit>> – real-time unit
- <<HwProcessor>> – hardware processor
- <<SchedulableResource>> – task/thread

Non-Functional Properties (NFPs)

Captures measurable attributes such as:

- Execution time
- Energy consumption
- Bandwidth
- Latency

These are critical in embedded systems.

Maps software components to hardware:

- Which task runs on which processor
- Memory allocation

MARTE allows transformation of UML models into:

- Queueing networks
- Scheduling models

Enables **early performance prediction**, crucial in engineering.

(A) Foundations

- Time
- Non-functional properties
- Generic resource modeling

(B) Design Model

- Hardware modeling
- Software modeling
- Allocation

(C) Analysis Model

- Performance analysis
- Schedulability analysis

Smart Traffic Signal System

Using MARTE, we model:

- Sensors (hardware resources)
- Controller logic (software tasks)
- Timing constraint (response within 100 ms)
- Allocation (controller on embedded processor)

MARTE vs Standard UML

Feature	UML	MARTE
Timing constraints	Limited	Explicit, formal
Hardware modeling	Weak	Strong
Performance analysis	Not supported	Supported
Embedded systems	Not specialized	Fully specialized

- Automotive (ECUs, ADAS systems)
- Aerospace (flight control systems)
- Telecommunications
- Industrial automation
- IoT systems

- MARTE is a bridge between design and analysis
- Enables model-driven engineering (MDE)
- Supports early validation of system performance

Limitations:

- Complexity
- Tooling limitations
- Steep learning curve

Official OMG specification:

<https://www.omg.org/spec/MARTE/>

MARTE is a UML extension that enables precise modeling, analysis, and validation of real-time and embedded systems by incorporating time, resources, and performance constraints into system design.